

# Consider how people use scientific explanations to meet a need or solve a problem

## Learning intention

- consider how people use scientific explanations to meet a need or solve a problem.

## Teach and model

- investigating how knowledge of magnetic force is used to sort metals in recycling, mining and food processing.
- examining how people use knowledge of friction to improve car or bicycle safety on slippery surfaces such as wet or icy roads.
- exploring how knowledge of the properties of plastic has influenced people to change how they purchase, use and dispose of plastic products.

## Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

## Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

## Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.